

Network Security Rubric — Answers

Healthcare IAM Operations Simulation

Network Topologies, Protocols, Security Fundamentals, Access Control, Wireless, Tools, and Incident Response

Project Name	Healthcare IAM Operations Simulation
Organization	Horizon Valley Medical Center (HVMC)
Rubric Covered	Network Security (7 Criteria)
Project Weeks	Week 1, Day 5 – Week 3, Day 4
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Lab Environment

Scenario: Horizon Valley Medical Center (HVMC) — a fictional community hospital (~30 employees, 10 departments) used to model realistic IAM and network security operations.

Domain: hvmc.local

Hypervisor / DC Platform: UTM running Ubuntu Server VMs (Mac host). HVMC-DC01 runs Samba4 as the Active Directory Domain Controller — Samba4 implements the same Kerberos/LDAP/DNS protocols as Windows AD DS.

Lab Subnet: 192.168.50.0/24 on a UTM virtual network (Shared Network mode, NAT-style outbound-only) named IAMLab-Net

Lab Hosts: HVMC-DC01 (192.168.50.10 — Ubuntu Server, Samba4 AD DC / DNS) • HVMC-CL01 (192.168.50.20 — domain-joined client) • HVMC-KALI (192.168.50.30 — scanning / IDS-IPS / SET / OpenCTI) • Gateway 192.168.50.1

Departments Referenced Below: Doctors, Nursing, Pharmacy, Laboratory, Human Resources, Finance, IT, Security, Reception, Contractors — matches the finalized Day 1 department list

1. Network Topologies — LAN

Single-segment LAN, star topology around one UTM virtual network switch (Shared Network mode), isolated from the home LAN/internet (NAT-style, outbound-only). Supports secure communications via: a single point of authentication (the Domain Controller), a contained blast radius (no inbound route in or out), simplified monitoring (one choke point for IDS/Wireshark), and centralized GPO/RBAC management.

2. Network Protocols and Architectures

OSI/TCP-IP mapped to CL01's Kerberos logon to DC01 (Layer 7 Kerberos/LDAP/DNS down to Layer 1 virtual NIC; TCP 88 for Kerberos, TCP 636 for LDAPS).

Subnetting: 192.168.50.0/24, mask 255.255.255.0, broadcast .255, usable range .1-.254, static allocations .1/.10/.20/.30, DHCP pool .100-.150

Secure Architecture: Kerberos (mutual authentication), LDAPS (encrypted directory queries), SMB signing (file share integrity), TLS 1.2+ (Entra ID cloud link)

3. Network Security Fundamentals

Firewall: Block-SSH-External inbound rule on DC01 (Ubuntu ufw/iptables), TCP 22, blocks all sources except 192.168.50.2 (admin workstation)

IDS: Suricata in detection mode on IAMLab-Net; local rule alerts on more than 15 SYN packets per 10 seconds toward DC01

IPS: Same signature class running in drop/inline mode via nfqueue

Detected Event: An Nmap SYN scan launched from Kali triggers the IDS alert (sid 1000010) against DC01:445

4. Access Control Measures

ACL on the Samba share \\HVMC-DC01\Nursing-Shared:

SG-Nursing-Shared-RW	→ Modify / Read-Write
SG-Doctors-Shared-Read	→ Read-Only
SG-Contractors-Shared-Deny	→ Explicit Deny
SG-IT-ITAdmin-Admin	→ Full Control

Model: DAC (owner-set Samba share permissions / POSIX ACLs) combined with RBAC (AD security groups assigned by role, via Samba4). MAC is explicitly noted as not applicable — Samba's POSIX ACL model, like Windows NTFS, is discretionary.

Standard User: SG-Nursing-Shared-RW — own-department read/write, no administrative rights

Privileged Admin: SG-IT-ITAdmin-Admin — full control plus delegated OU administration (via samba-tool), MFA required

5. Secure Wireless Networks

Demonstrated on the home Wi-Fi network used to manage the lab (the AD/Entra lab itself is wired-only).

SSID: HVMC-Lab-WiFi

Security: WPA3-Personal (WPA2/WPA3-transitional fallback), AES-CCMP/GCMP-256 encryption, 16+ character passphrase, isolated guest SSID

WIPS: Router-native rogue access point detection, or Kismet running on HVMC-KALI with a monitor-mode adapter

6. Network Security Tools

Wireshark: Capture on Kali's interface during a CL01 logon, filtered on kerberos || tcp.port==636; analysis notes an encrypted Kerberos payload alongside cleartext DNS resolution of hvmc.local

Vulnerability Scan: OpenVAS/Nessus Essentials run against DC01 and CL01 (findings: SMBv1, weak LDAP ciphers, missing patches, CVSS-classified)

Pentest Tool: nmap --script vuln -p 445,22 192.168.50.10, or Metasploit's smb_version fingerprinting module

7. Monitor and Respond to Network Security Events

Incident: a simulated brute-force attack (Hydra/Crowbar) from Kali against a decoy account on DC01 — triggering an IDS alert plus repeated failed-authentication entries in Samba's audit log (the Samba/Kerberos equivalent of Windows Event ID 4625), followed by an account-lockout log entry (equivalent of Event ID 4740). Samba logs authentication activity to its own logs (log.samba / auditd integration) rather than the Windows Security Event Log; the familiar Windows Event ID numbers are used here only for clarity, not as a literal log source.

Detect: IDS alert correlated with the repeated failed-logon audit entries, prioritized High

Contain: Block the source IP; confirm the account lockout

Eradicate: Verify there was no successful authentication in the same window (no 4624-equivalent entry)

Recover: Unlock the account; remove the temporary IP block

Lessons Learned: Add automatic, time-boxed blocking for repeated failed logons

Alert Prioritization: High — targets the DC or a privileged account, sustained · Medium — standard user, single source · Low — informational probe only

Full report with diagrams, tables, and rule blocks delivered separately as: Network_Security_Rubric_Report.pdf